

SolarTwin AI

Enterprise Solar Assessment Report

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Project Information

Parameter	Value
Location	Airoli, Airoli Station Road, Airoli Sector 2, Airoli, Navi Mumbai, Thane
Roof Area	12.36 m²
Panels	1
Capacity	0.55 kW
Orientation	Portrait

Weather

Parameter	Value
Temperature	29.7 °C
Cloud Cover	54 %
Wind	15.4 km/h
Humidity	76 %

Solar Production

Parameter	Value
Annual Energy	819 kWh
Daily Energy	2.2 kWh
Annual Saving	■ 6,555
Payback	5.0 Years
25 Year Profit	■ 130,876

AI Recommendations

- South-facing panels generally maximize solar generation.
- Consider subsidies or reducing installation cost to improve ROI.

Executive Conclusion

SolarTwin AI concludes that the analysed rooftop is suitable for a photovoltaic installation of approximately 0.55 kW. The estimated annual production is 819 kWh, with estimated annual savings of ■ 6,555. The generated layout maximizes the usable roof area while considering rooftop geometry and detected obstacles. The proposed design is recommended for further engineering validation before installation.